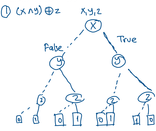
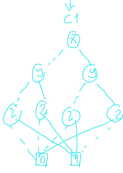
Logic - 5

Wednesday, 13 March 2024 15:24

① $(x \wedge y) \oplus z$ x,y,z



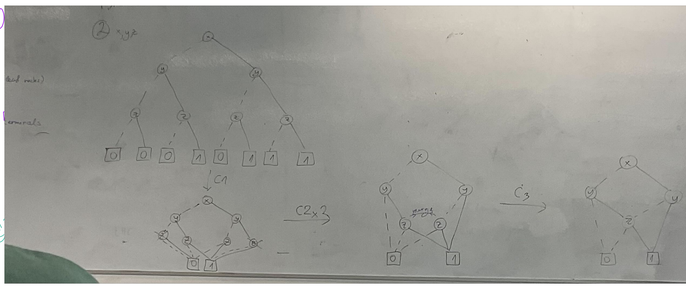
C1 → eliminating duplicate terminals
C2 → eliminating redundant
C3 → eliminating duplicate non-terminals



C3x2 →

C2 →

②



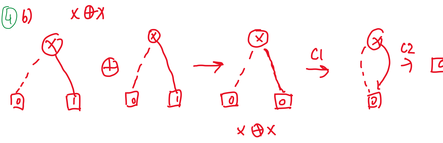
③

$x \vee \neg y \vee z$

x	y	z	ϕ
1	1	1	1
1	1	0	1
1	0	1	1
1	0	0	1
0	1	1	1
0	1	0	0
0	0	1	1
0	0	0	1

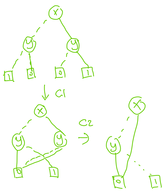
→ CNF $x \vee \neg y \vee z$

④ b) $x \oplus x$

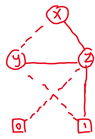


$x \oplus x$

⑤ a) $\neg(x \vee y)$

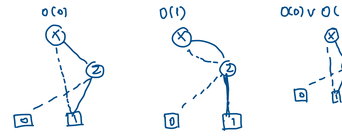


b) $\neg(x \vee y) \vee z$



c) $\exists y(\neg(x \vee y) \vee z)$

$y = 0$



∃ - there exits
∀ - for all

d) $\forall y(\neg(x \vee y) \vee z)$

$O(0) \wedge O(1)$

x=0	x=1	∃x ϕ	O(0) v O(1)
↓	↓		
O(0)	O(1)	∀x ϕ	O(0) ∧ O(1)

⑥ $\forall x\, \phi(x)$ only x occurs freely

↳ there are no other variables outside of the scope of quantifier

a) $\exists x\,(x \to y)$ is a tautology. HOLDS
if $x=0$ then $x \to y$ is always regardless of the value of y

b) Doesn't hold 'cause if $x=1$ $y=0$ then FALSE

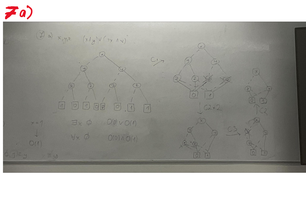
c) $\underbrace{\forall x\, \exists y\,(x \to y)}_{\text{tautology}} \equiv \underbrace{\forall x\,(x \to y)}_{y}$ - Doesn't hold

d) $\exists x\, \phi(x)$ $\underbrace{\text{tautology or contradiction}}_{\text{HOLDS}}$
if $x=0/1$ → either contradiction or tautology

e) $x \wedge \exists x\, \phi(x)$ is either tautology or contradiction DOESN'T HOLD
if $\exists x\, \phi(x)$ is contradiction → a contradicti
if $\exists x\, \phi(x)$ is tautology → $x \wedge \exists x\, \phi(x) \equiv x$

f) $\forall x\, \phi(x) \equiv \forall y\, \phi(y)$ HOLDS

the universal quantifier bounds all values of a variable regardless of its name

g) $\phi x \equiv \phi(y)$ DOESN'T HOLD

7 a)